

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Sri Ramakrishna Engineering College, India

Aug 2019 - May 2023

Bachelor of Engineering in Electronics and Communication

TECHNICAL SKILLS

Programming: C, Embedded C, C++, C#, Python, Java, SQL, Verilog

Embedded Systems: Firmware Development, Embedded Software, Bare-Metal Programming, Microcontrollers, Microprocessors, FPGA, SoC, FreeRTOS, RTOS, Real-Time Systems, Hardware-Software Integration, Hardware-Software Co-Design

Microcontrollers & Platforms: PIC16F887, ESP32, Arduino, Xilinx Arty, MicroBlaze

Embedded Interfaces & Peripherals: GPIO, UART, SPI, I2C, ADC, PWM, Timers, Interrupts, Peripheral Interfaces, AXI, Quad-SPI Flash

FPGA & Development Tools: Xilinx Vivado, Vitis, MicroBlaze, AXI IP Cores, Verilog, MPLAB IDE, Proteus

Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, Software Design Patterns, Multithreading, Concurrency, Memory Management, Linux, Ubuntu, SDLC

Testing & Debugging: Test Automation, Unit Testing, Verification & Validation, Functional Testing, Integration Testing, System Testing, Embedded Debugging, System-Level Debugging, Root Cause Analysis, Test Coverage Optimization

Tools & Practices: Git, Bitbucket, Jira, LabVIEW, NI TestStand, MAVIS, Agile, Scrum, Code Review, Technical Documentation

PROFESSIONAL EXPERIENCE

Project Engineer, Soliton Technologies, India

July 2022 – June 2024

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and test automation
- Implemented reusable software components and automated workflows using object-oriented design principles, improving development and testing efficiency
- Developed customized software solutions for Medtronic as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Integrated automated testing and validation workflows for system-level software, improving test coverage, reducing manual execution, and supporting repeatable software verification
- Collaborated with cross-functional engineering teams to debug system-level issues, perform root-cause analysis, troubleshoot application behavior, optimize workflows, and deliver production-ready software

Teaching Assistant, Department of Computer Science and Engineering, University of Arkansas

Sep 2024 – May 2026

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals
- Assisted students with control flow, functions, arrays, programming logic, debugging, and problem-solving techniques
- Supported Database Management Systems instruction for approximately 180 students, assisting with database concepts, assignments, assessments, and technical questions

PROJECTS

Autonomous Vehicle

Xilinx Arty FPGA, MicroBlaze SoC, Vivado, Vitis, AXI, Quad-SPI

- Designed and implemented an autonomous line-following vehicle using a Xilinx Arty FPGA and custom MicroBlaze-based SoC, integrating embedded hardware and software components
- Integrated IR line sensors, ultrasonic sensors, and motor drivers through AXI IP cores in Vivado for real-time sensor acquisition and actuator control
- Developed embedded firmware in C using Vitis for real-time line tracking, obstacle detection, sensor processing, and motor control
- Implemented Quad-SPI flash boot to enable autonomous system startup and embedded application execution
- Developed and debugged hardware-software interfaces across FPGA logic, MicroBlaze processing, sensors, and motor-control components

Expense Tracker with Predictive Analytics

C++, MySQL, Python, SQL, Database Integration

- Designed and implemented a cross-language expense management application integrating C++ business logic with a MySQL database for storing, retrieving, categorizing, and processing historical expense data
- Developed and optimized SQL queries to generate category-based expense summaries and reports, integrating database operations with the application's business logic
- Integrated Python-based analytics with the application workflow to analyze historical spending patterns and generate